君正®

T23 INDUS 开发板使用说明

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北京君正集成电路股份有限公司
Ingenic Semiconductor Co., Ltd.

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Release history

Date	Revision	Change
Jun.2023	1.0	First release

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1 使用环境准备

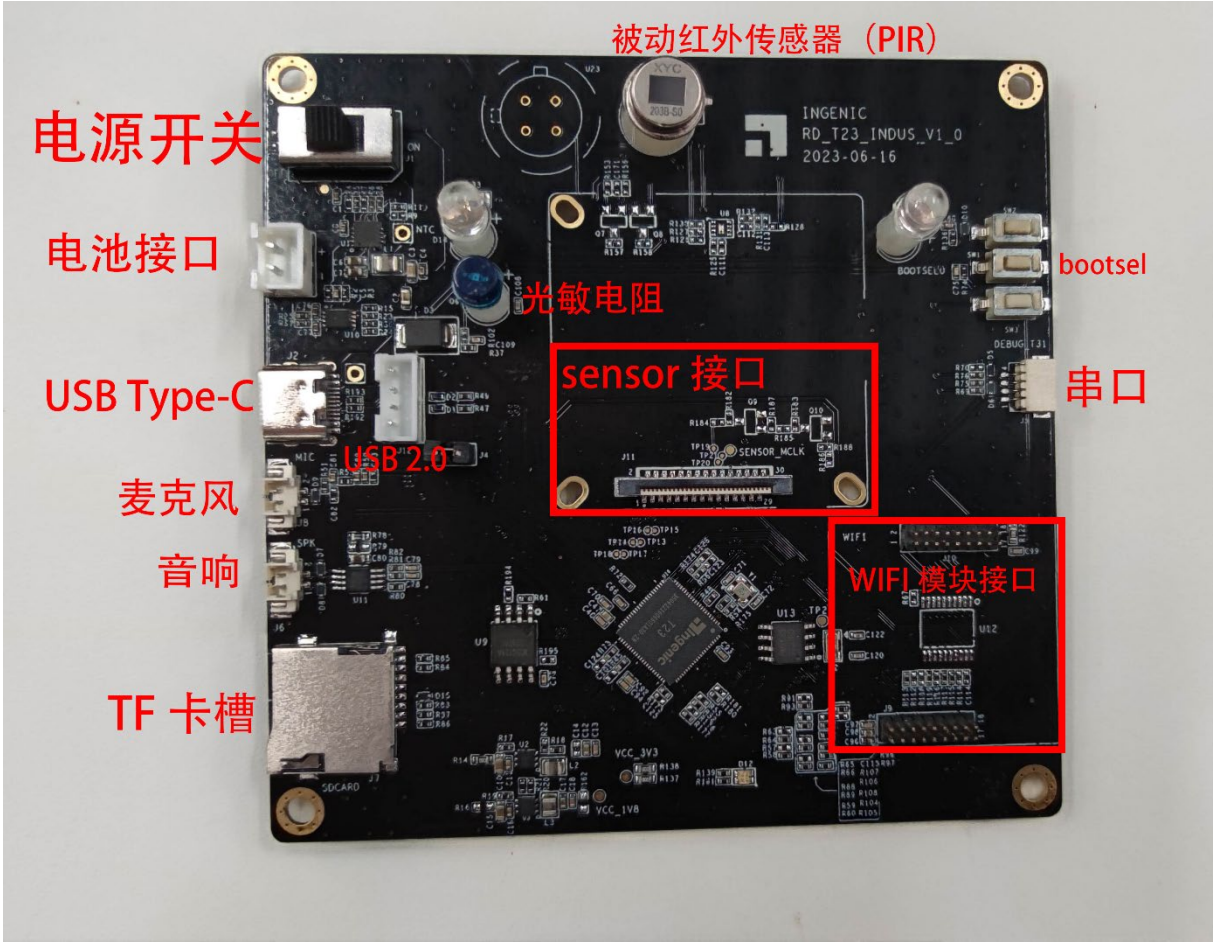

图 1.1　T23 开发板

1.1 设备连接

准备一台电脑，若是 windows 系统电脑，请安装 MobaXsterm 或者 secureCRT，若是 linux 系统电脑可以安装 minicom，用来连接串口，方便调试。请按照图 1.2 所示连接开发板。

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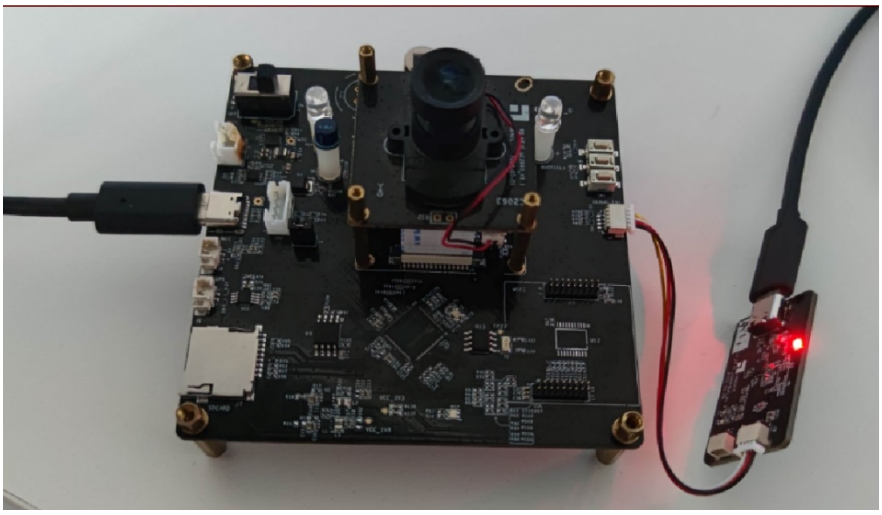

图 1.2 开发板接线

连接串口，这里以 Ubuntu 的 minicom 为例，在终端输入 sudo minicom –s 设置 115200 波特率，设置 Serial Device 为/dev/ttyUSB0(新串口板为 ttyACM0)，其他参数默认，保存退出即可进入串口。

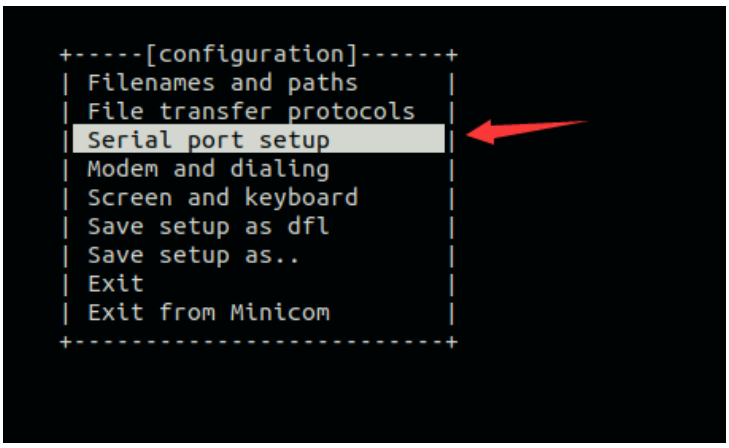

图 1.3 minicome 配置菜单

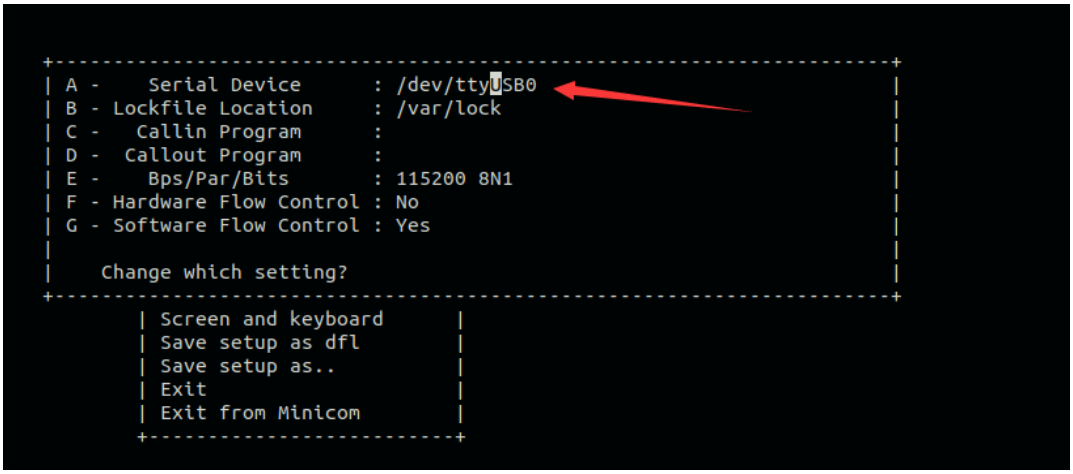

图 1.4 修改串口设备节点

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图 1.5 保存并退出设置

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2 网络配置

2.1 USB 转以太网说明

（1）USB 转以太网

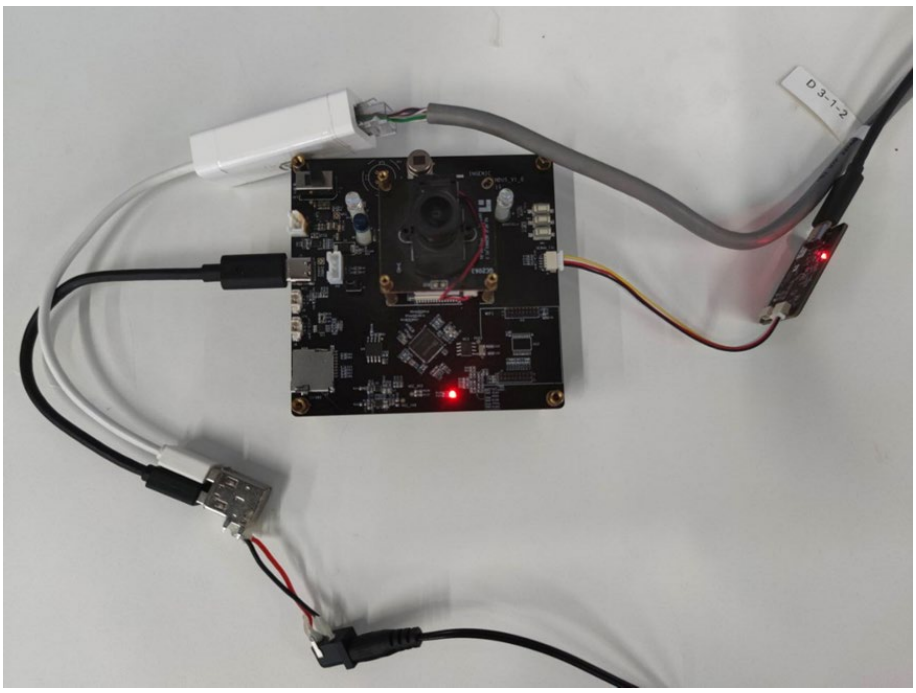

图 2.1 USB 转以太网接线

虽然开发板没有搭载 RJ45 以太网端口，但是我们可以采用 USB 转以太网的方式连接网络。

USB 转以太网的模块与开发板的接线如图 2.1 所示。

（2）T23 固件更新，可以使用以下几种方式：

A、 PC 端 USB 烧录工具

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B、 系统正常启动，挂载 SD 卡，flashcp 烧录
C、 系统无法启动，Bootsel 选择从 SD 卡启动，fatload 烧录。
具体方式，请参考 SDK 文档对应的烧录方式详细操作说明。

（3）编译开发板固件

如需使用 USB 转以太网，需要在编译内核时加入 USB 转驱动，首先使用

./build/build_camera_all.sh

编译整个固件，编译好后，
进入/out/camera 目录删除 uImage.jzlzma，进入 os/kernel 目录，执行

make zeratul_SOC_T23Z_camera_IIC_V1_defconfig

加载默认的内核配置
然后执行

make menuconfig

进入配置界面，如下配置

```
   a) Eth Config
   Device Drivers  --->
        [*] Network device support  --->
             USB Network Adapters  --->
                <*> Multi-purpose USB Networking Framework
                <*>   ASIX AX88xxx Based USB 2.0 Ethernet Adapters
(NEW)
   b) USB Mode Config
      Device Drivers  --->
          [*] USB support  --->
             <*> DesignWare USB2 DRD Core Support
                 Driver Mode (Host Mode Only)  --->
```

之后执行 make uImage 编译内核编译完成的内核在 os/kernel/arch/mips/boot 下，将编出来的 uImage.jzlzma 替换进/out/camera 中执行：

./build/build_camera_all.sh

生成新的固件，新固件中就包含 USB 转以太网驱动，将新的固件烧录到开发板。

2.2 T23 操作说明

给 T23 上电后，直接配置网络即可

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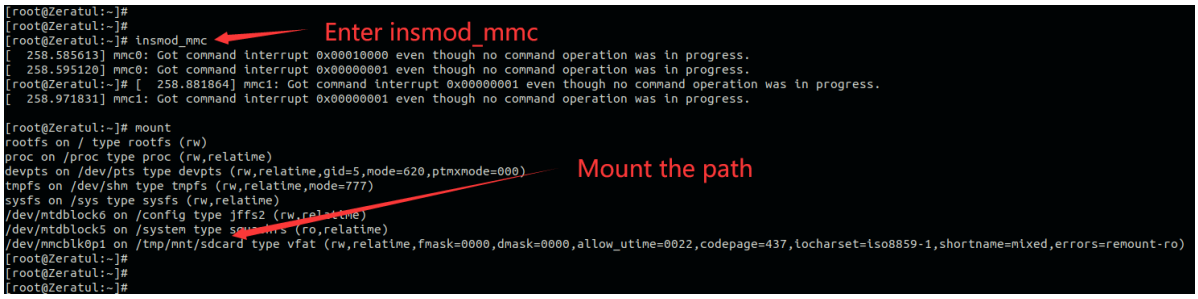
```
[root@Zeratul:~]#
[root@Zeratul:~]#
[root@Zeratul:~]# ifconfig eth0 10.3.32.233
[root@Zeratul:~]#  route add default gw 10.3.32.1
[root@Zeratul:~]# ifconfig
eth0      Link encap:Ethernet  HWaddr CE:B4:F8:0E:2F:E2
          inet addr:10.3.32.233  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

[root@Zeratul:~]#
[root@Zeratul:~]#
[root@Zeratul:~]#
[root@Zeratul:~]#
```

按照上图步骤操作，T23 网络可用。

2.3 挂载 SD 卡

输入 insmod_mmc 即可挂载 SD 卡，SD 卡挂载在/tmp/mnt/sdcard 目录下。




```
[root@Zeratul:~]#
[root@Zeratul:~]#
[root@Zeratul:~]# insmod_mmc
[  258.585613] mmc0: Got command interrupt 0x00010000 even though no command operation was in progress.
[  258.595120] mmc0: Got command interrupt 0x00000001 even though no command operation was in progress.
[root@Zeratul:~]# [  258.881864] mmc1: Got command interrupt 0x00000001 even though no command operation was in progress.
[  258.971831] mmc1: Got command interrupt 0x00000001 even though no command operation was in progress.

[root@Zeratul:~]# mount
rootfs on / type rootfs (rw)
proc on /proc type proc (rw,relatime)
devpts on /dev/pts type devpts (rw,relatime,gid=5,mode=620,ptmxmode=000)
tmpfs on /dev/shm type tmpfs (rw,relatime,mode=777)
sysfs on /sys type sysfs (rw,relatime)
/dev/mtdblock6 on /config type jffs2 (rw,relatime)
/dev/mtdblock5 on /system type squashfs (ro,relatime)
/dev/mmcblk0p1 on /tmp/mnt/sdcard type vfat (rw,relatime,fmask=0000,dmask=0000,allow_utime=0022,codepage=437,iocharset=iso8859-1,shortname=mixed,errors=remount-ro)
[root@Zeratul:~]#
[root@Zeratul:~]#
[root@Zeratul:~]#
```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3 设备出图

上电后可以执行 dmesg |grep TTFF，查看是否出流：

```
[root@Zeratul:~]# dmesg |grep TTFF
[    0.012119] TTFF tx_isp_riscv_hf_prepare:238 tx_isp_stop_riscv start:196
[    0.351168] TTFF tx_isp_riscv_hf_prepare:261 wait 8 frame done:535
[    0.431167] TTFF tx_isp_riscv_hf_prepare:291 tx_isp_stop_riscv done:615
[    0.442455] TTFF frame_channel_fast_start 109 W:2304 H:1296 N:2
[    0.501737] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:686
[    0.566043] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:750
[    0.663698] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:848
[    0.699449] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:884
[    0.769819] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:954
[    0.836983] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1021
[    0.901910] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1086
[    1.003414] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1188
[    1.032563] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1217
[    1.103733] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1288
[    1.175908] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1360
[    1.237257] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1421
[    1.302225] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1486
[    1.370312] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1555
[    1.433854] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1618
[    1.501136] TTFF isp_vic_interrupt_service_routine:898 Chn:0  Frame  Start:1685
[root@Zeratul:~]#
```

3.1 主进程运行

直接执行 sample，会在/tmp/目录下生成 stream-0.h264，执执行拷贝命令

```
cp stream-0.h264 /tmp/mnt/sdcard/
```

将视频拷贝到 SD 卡，执行 sync 即可。

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```
[root@Zeratul:202203]# ./sample-Encoder-video &
[root@Zeratul:202203]# [   19.896603] TTFF frame_channel_buffer_done:297 Chn:1 Buf:0 Write Done:20081
[frame_pooling_thread--419 Channel:1 ]:20088(ms)
[IMP_Encoder_GetStream_Impl--2533 Channel:1 ]:20095(ms)
streamNum[1]:FPS: 0.05,Bitrate: 2.38(kbps)
[   19.963256] TTFF frame_channel_buffer_done:297 Chn:1 Buf:1 Write Done:20147
[   20.029921] TTFF frame_channel_buffer_done:297 Chn:1 Buf:0 Write Done:20214
[   20.096597] TTFF frame_channel_buffer_done:297 Chn:1 Buf:1 Write Done:20281
[   20.163255] TTFF frame_channel_buffer_done:297 Chn:1 Buf:0 Write Done:20347
[   20.229923] TTFF frame_channel_buffer_done:297 Chn:1 Buf:1 Write Done:20414
[   20.296583] TTFF frame_channel_buffer_done:297 Chn:1 Buf:0 Write Done:20481
[   20.363255] TTFF frame_channel_buffer_done:297 Chn:1 Buf:1 Write Done:20547
[   20.429923] TTFF frame_channel_buffer_done:297 Chn:1 Buf:0 Write Done:20614
[   20.496590] TTFF frame_channel_buffer_done:297 Chn:1 Buf:1 Write Done:20681
[   20.563253] TTFF frame_channel_buffer_done:297 Chn:1 Buf:0 Write Done:20747
streamNum[1]:FPS: 15.06,Bitrate: 13.74(kbps)
streamNum[1]:FPS: 15.00,Bitrate: 41.60(kbps)
streamNum[1]:FPS: 15.00,Bitrate: 14.26(kbps)
streamNum[1]:FPS: 15.00,Bitrate: 39.85(kbps)
streamNum[1]:FPS: 15.00,Bitrate: 6.14(kbps)
streamNum[1]:FPS: 15.00,Bitrate: 28.42(kbps)

[root@Zeratul:202203]# cd /tmp/
[root@Zeratul:tmp]# cp stream-0.h265 /tmp/mnt/sdcard/
[root@Zeratul:tmp]#
[root@Zeratul:tmp]# sync
[root@Zeratul:tmp]#
```

Run sample

Copy video to SD card

sync

将视频拷出后可以在 windows 使用播放器播放也可以在 Ubuntu 安装 ffmpeg 执行 ffplay 查看：

```
ffplay stream-0.h264
```

把 carrie-server 可执行程序放在文件系统，手动运行./carrier-server --st=gc2063。运行结束会出现如下打印

```
Function usage:
        showfps: Show FPS and bitrate infomation
        osd: Switch on/off timestamp OSD
        snap: Snapshot a JPEG picture
        display: Display on LCD
Initializing...
---- FPGA board is ready ----
  Board UID : 30AB6E51
  Board HW ID : 72000760
  Board rev.  : A8A57397
  Board date  : 20200311
-----------------------------
...done initializing
> Timestamp OSD start
OSD start
Play this video stream using the URL:
        rtsp://192.168.1.10:8554/main
Play this video stream using the URL:
        rtsp://192.168.1.10:8554/second
Timestamp OSD start
OSD start
streamNum[1]:FPS: 24.92,Bitrate: 347.92(kbps)
```

```
Play this video stream using the URL:
    rtsp://192.168.1.10:8554/main
Play this video stream using the URL:
    rtsp://193.192.168.1.10:8554/second
```

可以通过两个 URL 播放主码流和次码流。

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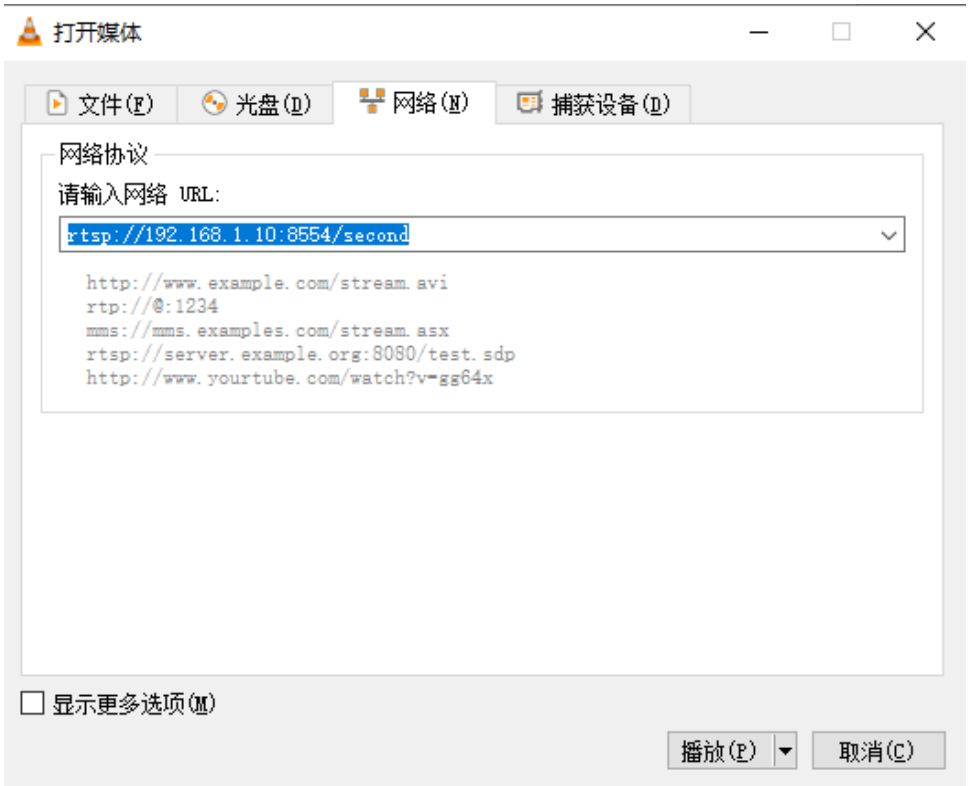
参数解释如下：
--st=gc2063 为指定第一个 sensor 型号

3.2 VLC 浏览视频

打开 VLC 输入链接地址 rtsp://192.168.1.10:8554/main，点击播放按钮。



3.3 通过 ubuntu 执行 ffplay 查看

```
ffplay rtsp://192.168.1.10:8554/main
ffplay rtsp://192.168.1.10:8554/second
```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